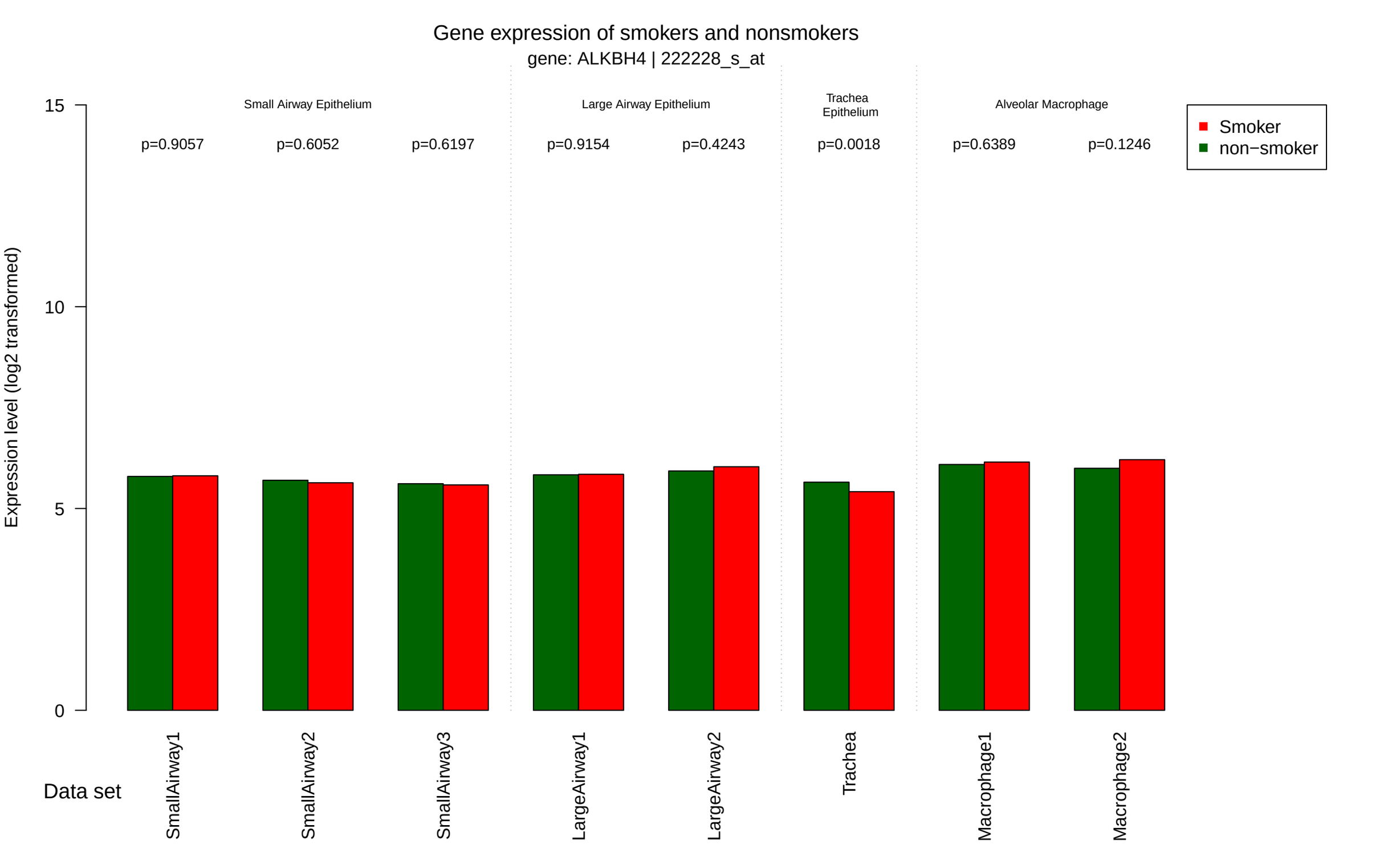
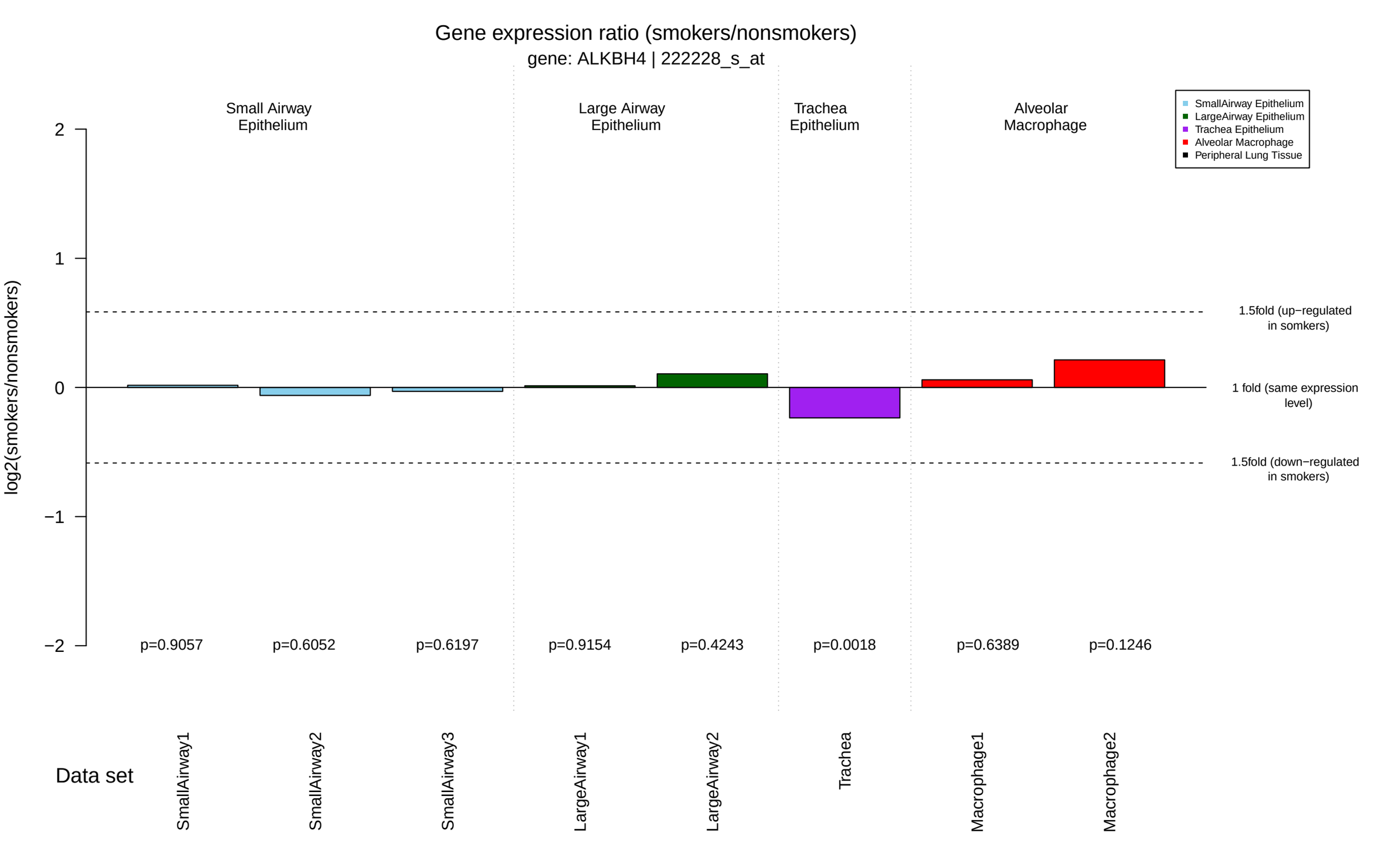
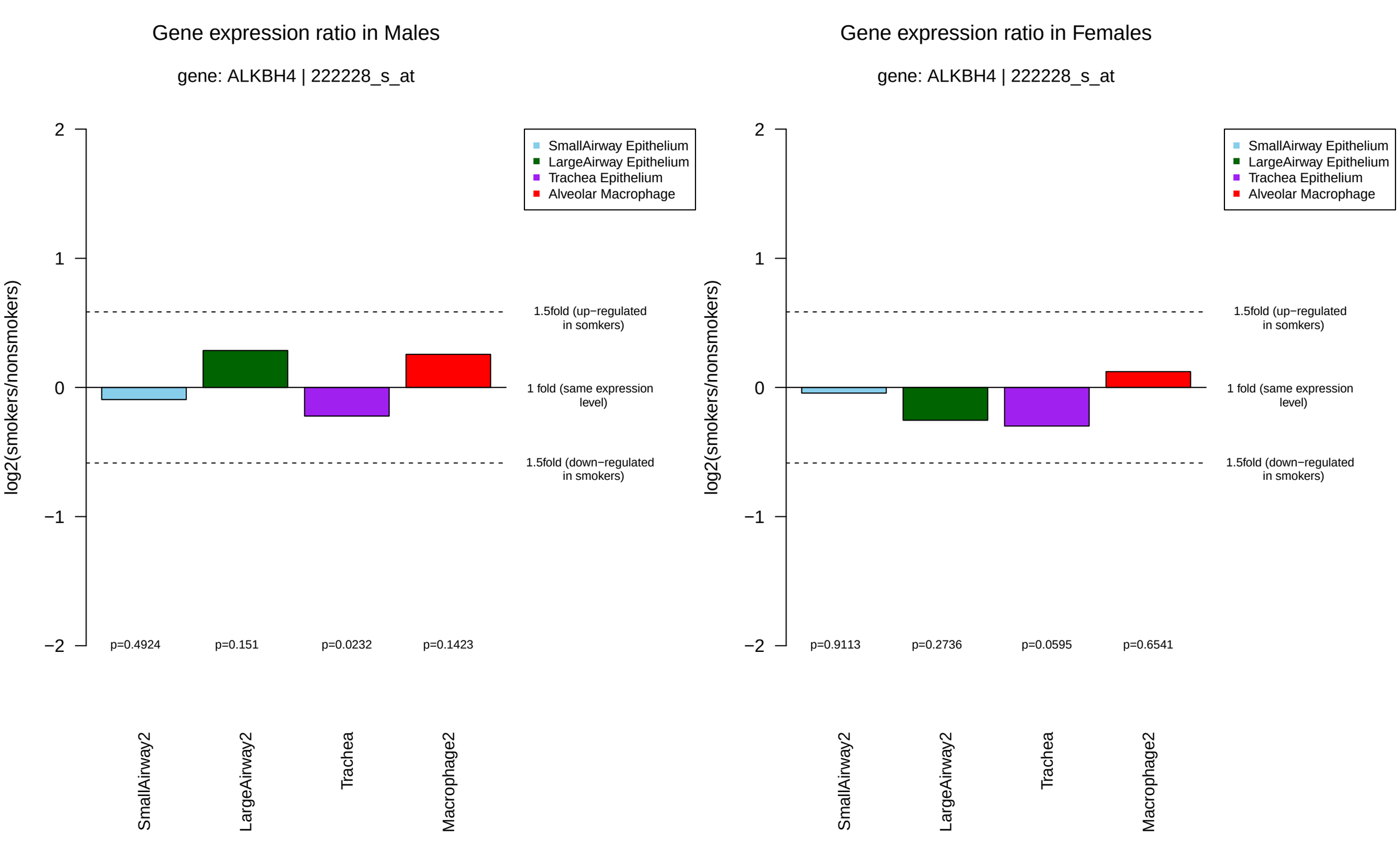
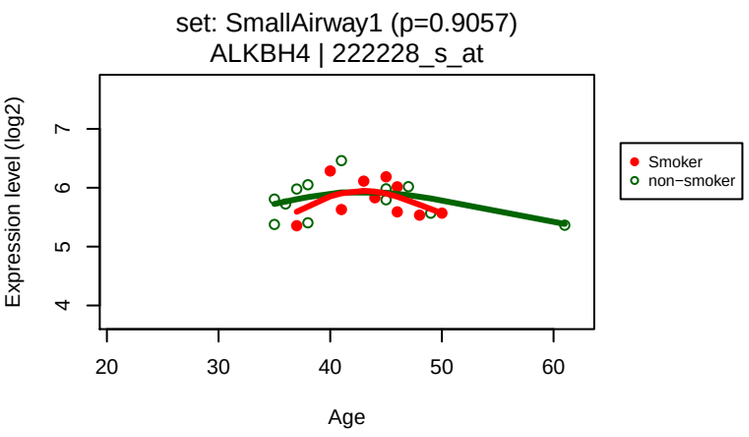
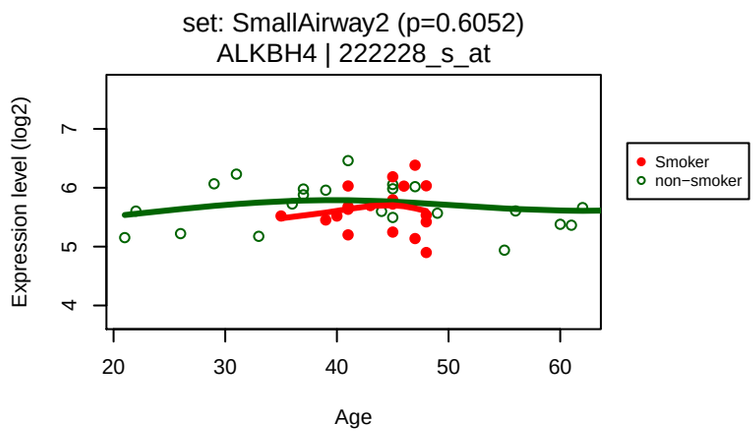
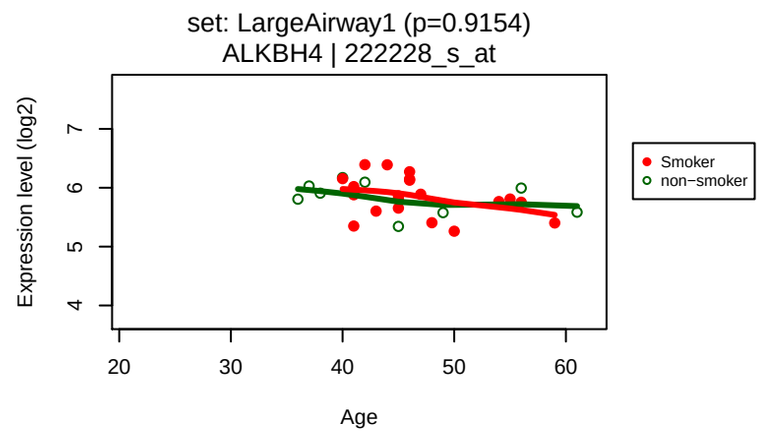

Gene expression of smokers and nonsmokers
gene: ALKBH4 | 222228_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9057
p=0.6052
p=0.6197
p=0.9154
p=0.4243
p=0.0018
p=0.6389
p=0.1246
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

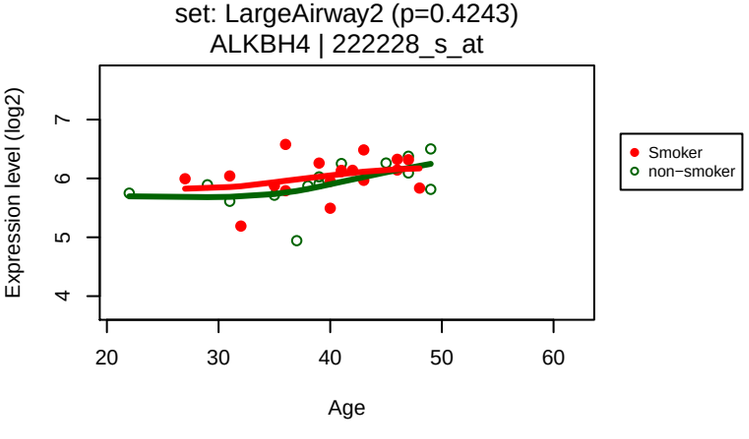

Gene expression ratio (smokers/nonsmokers)
gene: ALKBH4 | 222228_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9057
p=0.6052
p=0.6197
p=0.9154
p=0.4243
p=0.0018
p=0.6389
p=0.1246
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

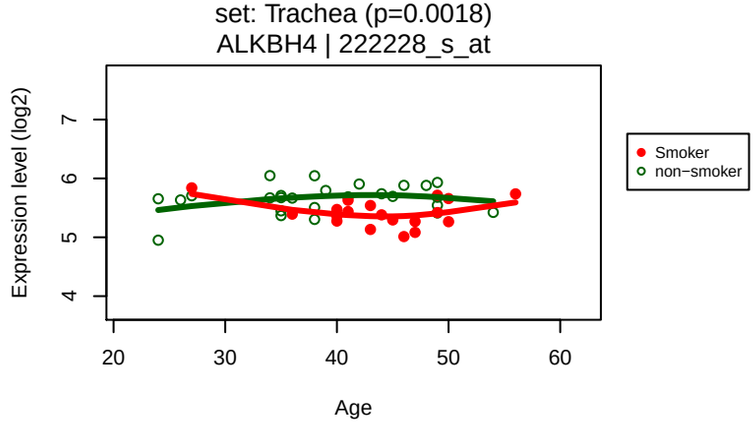

Gene expression ratio in Males
gene: ALKBH4 | 222228_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4924
p=0.151
p=0.0232
p=0.1423
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALKBH4 | 222228_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9113
p=0.2736
p=0.0595
p=0.6541
SmallAirway2
LargeAirway2
Trachea
Macrophage2

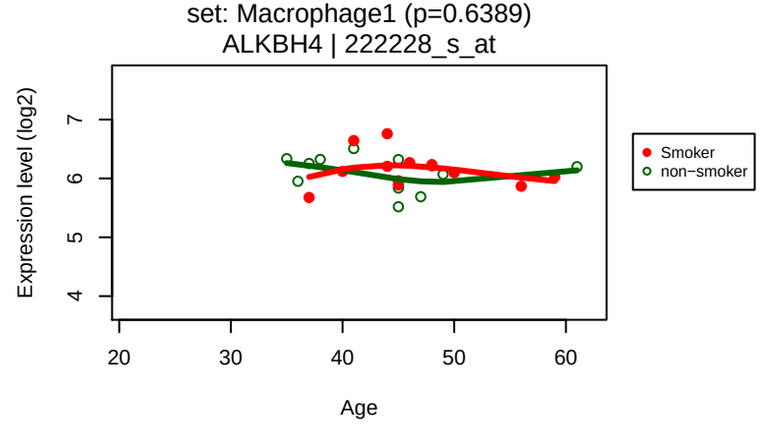

set: SmallAirway1 (p=0.9057)
ALKBH4 | 222228_s_at
Expression level (log2)
Age
Smoker
non−smoker

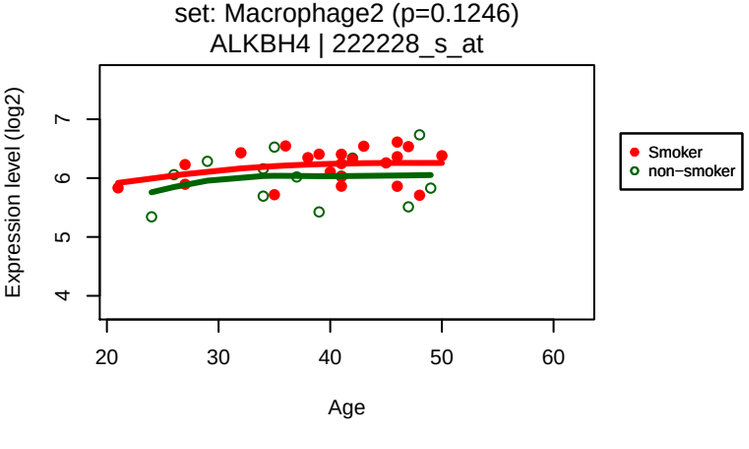

set: SmallAirway2 (p=0.6052)
ALKBH4 | 222228_s_at
Expression level (log2)
Age
Smoker
non−smoker

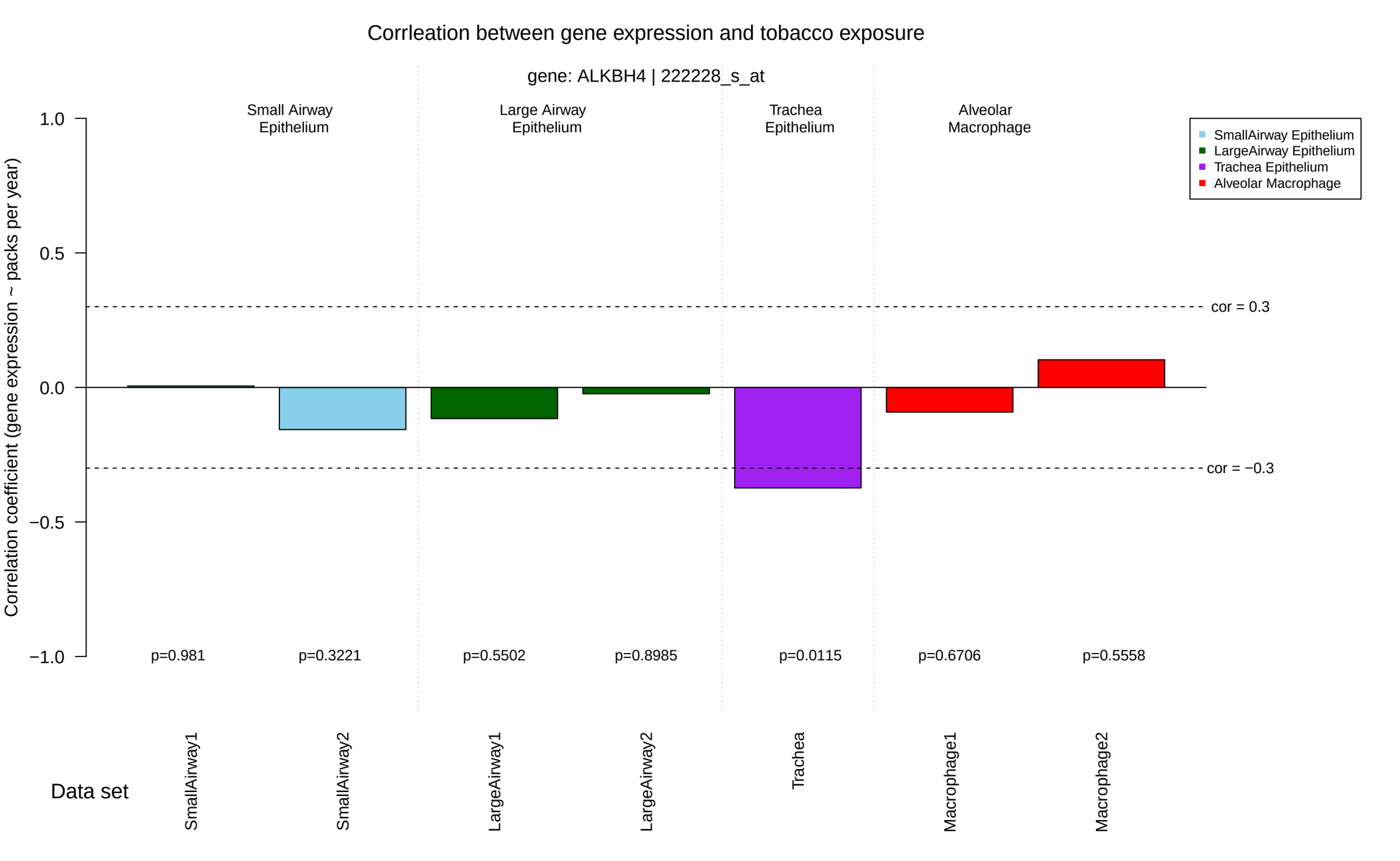

set: LargeAirway1 (p=0.9154)
ALKBH4 | 222228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4243)
ALKBH4 | 222228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0018)
ALKBH4 | 222228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6389)
ALKBH4 | 222228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1246)
ALKBH4 | 222228_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALKBH4 | 222228_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.981
p=0.3221
p=0.5502
p=0.8985
p=0.0115
p=0.6706
p=0.5558
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2